

Development of an Embedded HPEC Telemetry Unit and Windshield HUD for the TUKZIE Rev 0 Platform



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Submitted to the Department of Electrical Engineering at the University of Cape Town
in partial fulfilment of the academic requirements for a Bachelor of Science degree in
Electrical and Computer Engineering.

August 21, 2026

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Abstract

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Chapter 1

Introduction

1.1 Background and motivation

Rapid urbanisation is increasing demand for efficient urban transport and goods delivery. The United Nations projects that approximately 68% of the world's population will live in urban areas by 2050 [2], while road safety and transport emissions remain significant concerns [1]. These pressures have encouraged the use of lightweight electric vehicles for last-mile logistics, particularly where congestion, parking constraints, and operating costs limit conventional delivery vehicles [45].

The TUKZIE Flagship Programme seeks to develop an energy-aware, secure, and autonomous-ready urban freight platform. Its Rev 0 vehicle is a 72 V, 2200 W electric cargo trike intended as a testbed for telemetry and driver-assistance technologies. A study of medium-size deliveries in Dar es Salaam reported lower route-specific operating costs for an electric tricycle than for motorcycles, petrol tuk-tuks, and a light-duty car [3]. Although these results cannot be transferred directly to Cape Town, they support further development and field evaluation of locally appropriate electric freight vehicles.

Vehicle telemetry is useful only when sensing, processing, presentation, and communication operate as an integrated system. Inertial measurements must be interpreted in relation to vehicle speed, payload, mounting position, and powertrain vibration. Remote communication must tolerate outages, while the driver interface must avoid unnecessary visual demand. HUD research indicates that presentation location and information density affect attention and driving performance [5, 37]. NHTSA guidance recommends limiting individual visual-manual glances away from the roadway to two seconds [6], but

placing information in the forward view does not automatically make it safe.

High-Performance Embedded Computing (HPEC) and edge processing provide a means of timestamping sensor streams, extracting features, and maintaining local operation within limited power and timing budgets [28, 29]. This project therefore develops an integrated telemetry unit and removable windshield HUD for the TUKZIE Rev 0. The prototype is initially scoped toward Technology Readiness Level 5 through validation in controlled, relevant conditions. Progress toward Level 6 will be claimed only if the complete system is demonstrated reliably during representative vehicle operation [7].

1.2 Objectives of the study

1.2.1 Problem statement and research questions

The engineering problem is to acquire synchronised data on a vibrating, power-constrained vehicle; distinguish road-induced motion from vehicle-specific vibration; maintain bounded processing and display latency; and preserve telemetry during intermittent cellular connectivity. These functions must operate without obstructing the driver’s view or making local sensing and alerts dependent on a remote service [21].

The study addresses the following questions:

1. How repeatably can a low-cost edge system characterise vehicle-specific ride smoothness across repeated runs, speeds, payloads, and labelled road segments?
2. Which HUD information hierarchy and update strategy can present critical telemetry while limiting visual clutter and driver distraction?
3. What latency, data-loss, and recovery performance can be achieved across sensor acquisition, edge processing, local logging, and 4G/LTE synchronisation?
4. Which time- and frequency-domain features best separate road excitation from repeatable motor and powertrain vibration?
5. Can low-cost environmental and air-quality sensors provide complete, time-synchronised, spatially referenced data suitable for route-level anomaly screening?

1.2.2 Project objectives

The project will:

1. integrate a removable module containing IMU, environmental, positioning, power-train or BMS interface, storage, and cellular-communication subsystems;
2. implement a deterministic multithreaded pipeline that preserves timestamps, identifies missing samples, and computes interpretable ride features;
3. establish a vehicle-specific vibration baseline and evaluate ride-index repeatability under controlled operating conditions;
4. evaluate environmental-data completeness while treating the measurements as screening data rather than regulatory pollutant concentrations;
5. develop a removable HUD for speed, battery state, fleet notifications, and prioritised alerts; and
6. validate acquisition reliability, processing and display latency, local logging, and recovery after a communication outage.

The intended contribution is a measurable, open architecture rather than a claim of reference-grade pavement profiling or production-ready driver assistance. The resulting prototype should support later route comparison, vehicle characterisation, and fleet experiments on the TUKZIE platform.

1.3 Scope and limitations

The study is limited to the TUKZIE Rev 0 cargo trike manufactured by SA Motorcycles [20]. The system comprises a high-rate IMU path, lower-rate environmental and positioning sensors, local storage, edge processing, a removable projected HUD, and a 4G/LTE link. Sensor acquisition, processing, HUD rendering, storage, and networking are separated so that slow input/output operations do not block time-sensitive acquisition [21].

The following limitations bound the claims of the study:

- Results from one vehicle and a limited set of urban routes may not generalise to other vehicles, payload configurations, or road environments.
- Without a reference road profiler, the ride output cannot be validated as IRI or presented as absolute pavement roughness.
- Environmental and air-quality measurements are intended for relative comparison and anomaly screening, not regulatory compliance, exposure assessment, or definitive source identification.
- The HUD uses a removable projector and visor or combiner rather than an optically engineered laminated windshield [9].
- Cellular delay and availability remain dependent on the operator, route, signal conditions, and remote server.
- Testing will use controlled, low-risk routes and will not represent unrestricted production-road deployment.

1.4 Report structure

Chapter 2 reviews the literature and derives design requirements. Chapter 3 will describe the system architecture, component selection, software, experiments, and safety procedures. Chapters 4 and 5 will present and discuss the results. Chapters 6 and 7 will provide the conclusions and recommendations.

Chapter 2

Literature Review

2.1 Review scope

The proposed system combines ride sensing, embedded processing, vehicular telemetry, environmental monitoring, and a windshield HUD. Evidence is therefore assessed not only by reported accuracy, but also by sampling requirements, computational cost, sensitivity to vehicle configuration, latency, communication loss, and driver workload. This distinction is important because results obtained with passenger cars, smartphones, or laboratory simulators may not transfer directly to a lightweight cargo trike.

The review follows a system-level sequence. It first establishes the freight-vehicle context, then examines the sensing and signal-processing problem, and finally considers the computing, communication, and display subsystems required to act on those data. Each section identifies both transferable findings and conditions that prevent direct adoption. The resulting requirements are intended to guide component selection, software partitioning, and the validation plan rather than serve as a catalogue of available technologies.

2.2 Light electric freight vehicles and urban logistics

Light electric freight vehicles can reduce the energy, space, and direct emissions associated with urban delivery, but the benefit depends on the route, payload, depot location, energy source, and comparison vehicle. Sebwa et al. reported lower operating costs for an electric cargo tricycle on selected Dar es Salaam delivery routes, together with energy use of up

to 0.54 kWh for a one-way trip [3]. The study provides useful regional evidence, although its fuel prices, route choices, payload assumptions, and exclusion of electricity-generation emissions limit direct transfer to Cape Town.

Studies using broader system boundaries reach similarly conditional conclusions. Saenz et al. reported life-cycle greenhouse-gas reductions of 51–72% for a tricycle logistics service, while Choubassi et al. showed that cargo-cycle viability depends on labour, route structure, capacity, and depot location [4, 45]. Network-level analyses likewise found that delivery density, vehicle range, capacity, and the share of suitable deliveries determine the benefit of e-bikes and e-trikes [46, 47, 48]. The implication for TUKZIE is that later cost or sustainability claims require measured route, speed, payload, energy, and service data. The present project provides the telemetry architecture needed for those measurements, rather than attempting to prove a fleet-level benefit from one prototype.

The operational context also affects the telemetry design. A compact three-wheeled chassis is likely to respond differently from a passenger car when payload position, tyre pressure, speed, or road surface changes. Route and powertrain measurements should therefore be recorded with the IMU stream. This allows a later analysis to distinguish a rough route from a change in vehicle loading or operating state and prevents logistics conclusions from being separated from the physical conditions under which the data were collected.

2.3 Ride smoothness and inertial sensing

Road roughness and vehicle ride smoothness are related but are not interchangeable. The International Roughness Index (IRI) is derived from a measured longitudinal road profile using a standard quarter-car model [12]. A chassis-mounted accelerometer instead measures the combined response of the road, tyres, suspension, payload, speed, sensor orientation, and mounting location. Raw vertical acceleration therefore cannot be labelled as IRI without a validated mapping and suitable reference measurements.

Vehicle-response studies nevertheless show that inertial data can support roughness screening. González et al. classified road profiles from simulated axle and body accelerations, but the inferred result depended on the assumed vehicle model [13]. Smartphone reviews identify placement, orientation, speed, sampling rate, and inconsistent ground truth as persistent causes of variation [14]. Cycleway and lightweight-platform studies extend the method beyond passenger cars, using combinations of IMU, GPS, camera data,

and passenger ratings [42, 43, 44]. Their calibration remains platform-specific, which supports a TUKZIE-specific index rather than direct adoption of published thresholds.

Sampling rate must follow the measurement objective. Kalyankar-Narwade et al. used an 8 Hz vehicle-mounted IMU to reconstruct a scooter drive cycle and estimate energy consumption [60]. That rate captured relatively slow speed and acceleration changes, but does not establish suitability for road-impact or vibration analysis. Alatoom and Smadi’s embedded pavement framework is closer to the proposed microcontroller and single-board-computer architecture because it provides controlled mounting, sampling, and local processing [11]. The TUKZIE sampling rate must therefore be justified from the vibration bandwidth and verified through timestamp and missing-sample analysis.

An IMU signal contains gravity, orientation changes, vehicle-body motion, road excitation, motor vibration, transient impacts, and sensor noise. Orientation compensation and a vehicle-fixed coordinate frame are required before runs can be compared. Useful low-cost features include vertical RMS acceleration, standard deviation, jerk, robust peak measures, crest factor, and energy in selected frequency bands. Stationary motor-speed tests, coast-down tests, and repeated traversals can then identify components associated with the powertrain and those that change with the road surface.

Bui et al. demonstrated real-time rough-road processing on an edge microcontroller [22]. Yiliguogqi et al. used a compact SVM feature set at 10 Hz and reported 80–100% correct classification across their defined vehicles and test segments [23]. Those results do not establish expected TUKZIE performance. They also do not show that an SVM or XGBoost pipeline can run alongside 100 Hz acquisition, logging, HUD rendering, and networking without missed deadlines. Acquisition must receive the highest priority, while machine-learning inference should run asynchronously on complete windows or offline. Execution time, memory use, queue occupancy, and sample loss must be measured before any model is retained for real-time operation.

Time-domain and frequency-domain analysis provide complementary information. RMS and robust peak measures describe the magnitude and occurrence of impacts, while spectral energy can reveal narrow-band motor components that persist on different road surfaces. The same spatial road feature produces a different temporal excitation when speed changes, so windows should be grouped or normalised by speed. Stationary tests at several motor speeds establish a powertrain baseline, coast-down tests reduce powered traction effects, and repeated traversals show which features recur at the same route location. This staged procedure is more defensible than selecting one threshold from a different vehicle.

The project should consequently report a repeatable, vehicle-specific ride index and event log. Agreement across repeated runs, sensitivity to speed and payload, and comparison with a reference IMU or labelled route are defensible validation criteria. ISO 2631-1 may inform a separate comfort-oriented measure, but it does not convert chassis acceleration into IRI [15].

2.4 Mobile environmental sensing

Low-cost environmental sensors can increase spatial coverage, but their output is affected by temperature, humidity, cross-sensitivity, ageing, calibration, and deployment conditions [56]. Mobile use adds airflow, speed, orientation, and rapidly changing background conditions. DeSouza et al. found that stationary correction models did not transfer uniformly to mobile PM_{2.5} measurements, although repeated mobile coverage could still identify persistent hotspots [57].

Ramos-Sorroche et al. provide a useful architectural comparison through a vehicle system combining environmental sensors, on-board processing, a single-board computer, and 4G and LoRaWAN links [61]. Their results support local preprocessing, georeferenced records, and communication-quality logging, but their dedicated gas-sensing units cannot be equated with a compact integrated sensor. The TUKZIE payload should therefore be treated as a route-screening instrument. Records should include location, time, warm-up and quality flags, while candidate anomalies should be checked for recurrence. Without reference calibration, the system cannot make regulatory or source-identification claims.

Sensor placement is part of the measurement method. An environmental inlet inside a sealed electronics enclosure would measure enclosure conditions rather than representative ambient air, while a position close to heat sources or disturbed airflow could introduce systematic bias. The payload must receive airflow without compromising weather protection. Initial evaluation should include warm-up behaviour, stationary co-location, missing-data checks, and repeated passage through the same areas. These tests do not create a regulatory instrument, but they can establish whether the mobile data are sufficiently stable for the screening objective stated in the research questions.

2.5 Edge computing and telemetry architecture

HPEC combines adequate throughput, predictable timing, and energy efficiency within constrained hardware [28, 29]. Edge computing places processing near the sensors, reducing response time, bandwidth use, and dependence on a remote service [16]. For TUKZIE, this means that acquisition, ride-event detection, logging, and the HUD must remain operational when cellular coverage is absent.

Embedded IoT surveys emphasise that communication, sensing, and computation compete for memory, processor time, and energy [30]. Edge-AI reviews add that model selection cannot be separated from the capabilities of the target hardware [31]. These findings support a two-level implementation. The ESP32-S3 can perform deterministic sensor acquisition and basic window preparation, while the Raspberry Pi can handle storage, higher-level processing, networking, and HUD rendering. The serial or network interface between them must include framing, timestamps, sequence numbers, and an explicit response to buffer overflow.

The relevant performance measure is predictable completion rather than maximum throughput. A producer-consumer design can separate high-rate acquisition from feature extraction, storage, display, and networking. Acquisition should timestamp samples and write to a bounded buffer; processing should consume complete windows; the HUD should read the latest validated state; and networking should transmit queued summaries asynchronously. Shared locks, blocking input/output, unbounded queues, and dynamic allocation can still create jitter even when separate threads are used. Performance should therefore be reported as distributions of sampling jitter, processing time, display age, queue occupancy, and end-to-end latency [21].

Embedded telemetry studies support filtering and event extraction close to the data source [41]. Guidance on embedded high-performance implementation similarly favours selecting the simplest platform and optimisation method that satisfies the timing budget [40]. This is relevant because an accelerator or complex classifier may improve average throughput without improving worst-case acquisition timing. The proposed system should first establish a measured schedule for acquisition, buffering, feature extraction, display updates, and network transfer. Optimisation is justified only where profiling identifies a real bottleneck.

Pérez-González et al. used ESP32 acquisition, Raspberry Pi edge processing, local visualisation, and LoRaWAN in an electric-racing telemetry system [24]. The division between

deterministic acquisition and higher-level processing supports the proposed ESP32-to-Raspberry-Pi architecture. Its reported delivery ratio and latency remain specific to the test track, gateway, and radio configuration, so local logging and asynchronous communication are still required.

If the unit reads an existing BMS or vehicle controller, the interface must be treated as a separate system boundary. Automotive networking literature shows a progression from CAN to higher-bandwidth Ethernet, with associated timing, interoperability, and security requirements [33, 49]. For the prototype, the practical requirements are to verify the available BMS interface, validate incoming data, authenticate remote endpoints, and ensure that external connectivity cannot alter local safety-relevant operation. Protocol surveys and gateway studies identify spoofing, denial-of-service, malformed messages, and insecure external interfaces as recurring risks [34, 35, 32, 36]. The initial design should therefore make remote commands non-critical, restrict exposed services, and fail safely when BMS or network data become stale.

2.6 HUD design and visual workload

HUDs reduce large downward eye movements, but they do not eliminate distraction. Studies by Liu, and by Liu and Wen, found that HUD and head-down interfaces produce different attention and driving-performance outcomes [5, 37]. Smith et al. further showed that both driving performance and the secondary display task must be assessed [18]. The TUKZIE interface should therefore minimise information rather than use the windshield as additional dashboard space.

A HUD comprises an image source, an optical or projection path, a combiner, and a computing unit that determines what is presented [10]. This separation is useful for the prototype because the display software, projector, and visor can be tested independently before vehicle integration. It also clarifies that a reflective film or visor is not a complete HUD. The final image depends on projector focus, surface geometry, ambient light, rider eye position, and the stability of the mechanical mount.

Display position and context are central. Full-windshield cues can direct attention toward the forward view, but simulator results do not show that arbitrary content is safe [50]. HUD imagery positioned farther from the driver’s forward view has been associated with poorer lane keeping and more unacceptable driving behaviour [51]. Windshield-display research consequently treats location, visual depth, contrast, content, interaction, and

the external scene as linked design variables [27]. Peripheral or misaligned graphics can compete with hazards for attention [8], while background complexity and interface opacity affect visual cognition [38].

Network-derived warnings require an additional validity rule. Charissis et al. demonstrated full-windshield presentation of network and low-visibility information [52], but the proposed system uses cellular telemetry rather than an assumed low-latency vehicular network. Fleet notifications must therefore carry timestamps and expire when stale. Immediate alerts should depend on local measurements wherever possible, and loss of the remote connection should remove or mark affected information rather than leave an old warning on the visor.

The removable visor creates a different optical problem from a production laminated HUD windshield. Production systems may use wedge-shaped PVB to reduce double reflections for a specific projector and eye position [9, 26]. TUKZIE will instead use a clear visor, reflective-film region, or separate combiner. The relevant tests are ambient-light contrast, focus distance, projector throw distance, eye-box size, ghost separation, and resistance to alignment changes under vibration. The mounting should remain adjustable and removable rather than copy a fixed automotive wedge angle.

Holographic optical elements and adaptive antiglare systems illustrate possible future solutions to contrast and glare [53]. Their materials, optical alignment, and manufacturing requirements are outside this project. The transferable point is that ambient light and glare change across the viewing region and throughout a route. The first prototype should address this with measured brightness control, a projector hood, careful combiner placement, and testing in representative daylight and night conditions before more complex optical materials are considered.

Display latency is also important because delayed graphics can disagree with the external scene. The transferable measurement principle is to time the complete path from sensor change to visible HUD update [39]. Complexity must be bounded as well. HUD studies show increased workload or changed user perception when too much information is presented [19, 17]. Park and Im recommended fewer than six simultaneous symbols in their study [25]. This is an initial design ceiling rather than a universal safety threshold.

The prototype should therefore maintain a stable hierarchy: persistent speed and battery state, brief prioritised alerts, and fleet or navigation information only when required. Invalid or stale values should be removed or clearly marked. Evaluation should include rendering time, display age, contrast under representative lighting, visible image area,

missed events, task performance, and perceived workload. The occlusion method can support controlled visual-demand assessment before low-risk vehicle trials [6, 54]. Broader automotive-HMI reviews also recommend considering usability, safety, user experience, and system security together rather than optimising visual appearance alone [55].

2.7 Cellular synchronisation and data integrity

Cellular delay and coverage are outside the embedded controller’s direct control, so remote synchronisation must be separated from the real-time path. Every stored record should include a monotonic timestamp, sequence number, session identifier, and quality flags. A store-and-forward queue and bounded retry policy are required to recover from outages without delaying local sensing or the HUD.

South African broadband measurements show that performance depends on both the radio link and the route to the remote service [58]. That study predates current LTE deployment and cannot define a 2026 Cape Town target. Vehicular LTE measurements also show that connectivity varies between urban and lower-coverage environments [59]. A route-specific baseline is therefore required.

Figueiredo et al. provide a useful protocol comparator from a connected e-bike using MQTT over a SIM7600 4G modem. Mean publish-subscribe times were 136.46 ms for QoS 0, 143.70 ms for QoS 1, and 224.60 ms for QoS 2; TLS added approximately 11.5%, and a national broker reduced latency by approximately 15.9% [62]. These values are not South African guarantees, but they support testing QoS 1 with TLS and locating the broker close to the operating network.

Evaluation should distinguish local processing latency from network synchronisation delay. Relevant measures are acknowledgement round-trip time, delivery ratio, outage duration, reconnect time, backlog size, backlog-clearance time, and modem indicators such as RSRP, RSRQ, and SINR. MQTT is suitable for compact event and status messages, while HTTP can be used for configuration or bulk transfer. Protocol choice alone does not provide reliability; recovery behaviour must be tested explicitly.

The communication experiment should include both naturally varying coverage and controlled outages. A shielded or disconnected-modem test can measure queue growth and deterministic recovery, while urban route tests capture operator and mobility effects. Reporting only average latency would conceal long delays that create stale displays or

growing backlogs. Median, percentile, and maximum observed values should be accompanied by the message rate, payload size, QoS, encryption state, broker location, and radio-quality indicators so that the results can be reproduced and compared with the literature.

2.8 Synthesis and design requirements

The literature supports low-cost inertial screening, local edge processing, mobile environmental sensing, asynchronous telemetry, and sparse HUD presentation. It also shows that each subsystem is sensitive to its deployment context. Acceleration features depend on vehicle configuration and speed; mobile environmental measurements require bounded interpretation; cellular performance varies by route and server; and HUD benefits diminish when lighting, vibration, location, and workload are ignored.

No reviewed study integrates these concerns for a removable system on a lightweight electric cargo trike. The project therefore contributes an architecture whose ride-smoothness repeatability, timing, data integrity, environmental-data completeness, and HUD presentation can be evaluated together. The review yields the following requirements:

1. Report a vehicle-specific ride index, not IRI, unless reference calibration is available.
2. Timestamp IMU data, account for orientation and speed, and compare powered, stationary, coast-down, and repeated-route conditions.
3. Keep acquisition, processing, display, storage, and networking non-blocking and measure timing distributions and data loss.
4. Keep machine-learning inference outside the acquisition-critical path unless measured timing proves that the sampling deadline is preserved.
5. Treat environmental measurements as screening data with quality flags.
6. Maintain local operation during cellular outages and verify store-and-forward recovery.
7. Evaluate the actual removable HUD under representative lighting, eye positions, background complexity, and vibration.
8. Link validation measures directly to the research questions and project deliverables.

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